Mohanaprasath Subramaniyan

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Data Engineer (2+ Years) • M.S. Data Science • Python, SQL & Scala • ETL/ELT Pipelines • Azure Databricks (Spark) • Data Modeling (Star/Snowflake Schema)

Technical Skills

Languages & Frameworks: Python (Pandas, PySpark), SQL (T-SQL, PL/SQL), Scala (basic), Bash **Data Engineering:** ETL/ELT Pipeline Design, Data Modeling (Medallion Architecture), Delta Lake, Stream Processing

Cloud & Platforms: Azure Databricks, Azure Data Factory, Snowflake, AWS (S3, EC2), Docker, Kubernetes (basic)

Storage & Databases: PostgreSQL, Azure SQL, MongoDB, Qdrant (Vector DB), ChromaDB, Redis

Tools: Airflow (Orchestration), Git, CI/CD, Power BI (Data Visualization), dbt (basic), Kafka

Experience

Data Engineer | **MODEC (Client)** *India, Jan 2024 – Jul 2024*
HESC (Employer)

- Designed and optimized **ETL/ELT pipelines** in Azure Databricks using **PySpark**, processing multi-terabyte FPSO sensor data into a Medallion architecture (Bronze/Silver/Gold).
- Engineered a **robust data modeling** strategy for predictive maintenance, reducing data redundancy by 30% and improving query performance for downstream analytics.
- Automated data quality frameworks using Python to validate schema consistency and handle late-arriving data, reducing reporting discrepancies by 40%.
- Built a scalable **semantic layer** in Power BI, enabling real-time drill-down capabilities for critical safety and asset utilization KPIs.

Data Engineer | **Equinor (Client)** *India, Aug 2023 – Dec 2023*
HESC (Employer)

- Consolidated disparate production and maintenance data streams into a centralized **Snowflake data warehouse**, ensuring a "single source of truth" for operational KPIs.
- Developed **automated validation scripts** and reconciliation logic to ensure high data integrity for business-critical audit logs.
- Implemented anomaly-detection monitoring for data ingestion pipelines, reducing pipeline downtime and improving response time by 20%.
- Optimized SQL queries and indexing strategies, leading to a 25% improvement in dashboard refresh rates.

Associate Data Engineer *India, Feb 2022 – Jun 2023*
NTT DATA

- Developed end-to-end **data ingestion workflows** from EMR systems into secure analytical environments, ensuring HIPAA compliance and data masking.
- Architected a customer-centric **data mart** that supported churn analysis, improving data accessibility for marketing analysts by 50%.
- Built and maintained **automated demand-forecasting pipelines** that integrated inventory logs with real-time supply chain signals.
- Collaborated with DevOps to containerize data processing tasks using **Docker**, ensuring environment parity across development and production environments.

Projects

Real-Time E-Commerce Analytics Pipeline (Streaming & CDC) *Kafka, Spark Streaming, PostgreSQL, Docker*

- Architected a **streaming data pipeline** using **Apache Kafka** to ingest clickstream data, achieving a throughput of 5,000+ events per second with minimal lag.
- Implemented windowed aggregations in **Spark Streaming** to calculate real-time conversion rates and inventory levels, sinking processed data into a **PostgreSQL** layer.
- Leveraged **Docker Compose** to containerize the entire ecosystem (Zookeeper, Kafka Broker, Spark Worker), ensuring reproducible deployment and scalability.

PolicyNexus - Agentic Data Orchestration Platform *Python, LangGraph, Docker, Qdrant, Airflow*

- Built a **multi-agent data pipeline** using LangGraph that autonomously orchestrates data retrieval from unstructured PDF stores and live APIs.
- Implemented a **hybrid search architecture** combining vector embeddings (Qdrant) and keyword search to optimize retrieval latency to sub-second levels.

Cloud Data Warehouse for Retail *PySpark, Delta Lake, Azure*

Databricks, SQL

- Engineered a scalable **Delta Lake** solution to handle high-velocity fashion retail data, implementing schema enforcement and time-travel features for auditing.
- Developed **dbt-style transformations** in Spark SQL to create aggregated business views, improving recommendation engine accuracy for a simulated e-commerce platform.

Education

Pace University, Seidenberg School of CSIS *M.S. in Data Science, GPA: 3.8* — *New York City, Sept 2024 – Jan 2026*

SNS College of Technology *B.E. in Mechanical Engineering, GPA: 8.6* — *India, 2020*